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Faces of Polytopes

In this lecture we will discuss faces, and the face lattice. Here we restrict our attention entirely to polytopes, although nearly everything can quite easily be generalized to polyhedra; see the exercises.

I hope that the reader enjoys the ease with which we will get the results in this lecture. In fact, nearly all results are “geometrically clear,” and as far as we need algebra to verify them, we get by with straightforward computations and the Farkas lemmas.

2.1 Vertices, Faces, and Facets

Definition 2.1. Let $P \subseteq \mathbb{R}^d$ be a convex polytope. A linear inequality $\mathbf{c}\mathbf{x} \leq c_0$ is *valid* for P if it is satisfied for all points $\mathbf{x} \in P$. A *face* of P is any set of the form

$$F = P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}$$

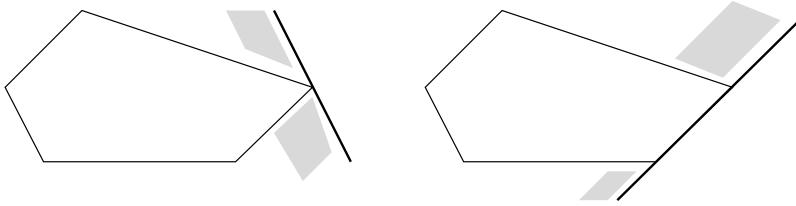
where $\mathbf{c}\mathbf{x} \leq c_0$ is a valid inequality for P . The *dimension* of a face is the dimension of its affine hull: $\dim(F) := \dim(\text{aff}(F))$.

For the valid inequality $\mathbf{0}\mathbf{x} \leq 0$, we get that P itself is a face of P . All other faces of P , satisfying $F \subset P$, are called *proper* faces.

For the inequality $\mathbf{0}\mathbf{x} \leq 1$, we see that $\emptyset$ is always a face of P .

The faces of dimensions 0, 1, $\dim(P) - 2$, and $\dim(P) - 1$ are called *vertices*, *edges*, *ridges*, and *facets*, respectively. Thus, in particular, the vertices are the minimal nonempty faces, and the facets are the maximal proper faces. The set of all vertices of P , the *vertex set*, will be denoted by $\text{vert}(P)$.

The following sketches show two valid inequalities for a 2-polytope; they define a vertex and an edge, respectively.



In the following two propositions we collect some simple but basic facts about faces.

Proposition 2.2. *Let $P \subseteq \mathbb{R}^d$ be a polytope.*

- (i) *Every polytope is the convex hull of its vertices: $P = \text{conv}(\text{vert}(P))$.*
- (ii) *If a polytope can be written as the convex hull of a finite point set, then the set contains all the vertices of the polytope: $P = \text{conv}(V)$ implies that $\text{vert}(P) \subseteq V$.*

Proof. Let $P = \text{conv}(V)$. Now if any vector $\mathbf{v}_i \in V$ can be written as a convex combination of the other vectors in V , then we can clearly substitute that representation into any convex combination of vectors in V , and thus get a smaller set of points $V' := V \setminus \mathbf{v}_i$, whose convex hull is $\text{conv}(V') = P$.

Now we **claim** that if $\mathbf{v}_i$ cannot be expressed as a convex combination of $V' = V \setminus \mathbf{v}_i$, then it is a vertex of P . Using Farkas lemma II (Proposition 1.8), we get

$$\begin{aligned}
 \mathbf{v}_i \notin \text{conv}(V') &\iff \nexists \mathbf{t} \geq \mathbf{0} : \mathbf{v}_i = V' \mathbf{t}, \mathbf{1} \mathbf{t} = 1 \\
 &\iff \nexists \mathbf{t} \geq \mathbf{0} : \begin{pmatrix} \mathbf{1} \\ V' \end{pmatrix} \mathbf{t} = \begin{pmatrix} 1 \\ \mathbf{v}_i \end{pmatrix} \\
 &\stackrel{\text{FL II}}{\iff} \exists \mathbf{a} : \mathbf{a} \begin{pmatrix} \mathbf{1} \\ V' \end{pmatrix} \geq \mathbf{0}, \mathbf{a} \begin{pmatrix} 1 \\ \mathbf{v}_i \end{pmatrix} < 0 \\
 &\iff \exists (\beta, -\mathbf{b}) = \mathbf{a} : \mathbf{b} V' \leq \beta \mathbf{1}, \mathbf{b} \mathbf{v}_i > \beta \\
 &\iff \exists \beta, \mathbf{b} : \mathbf{b} \mathbf{v}_j \leq \beta \text{ for } j \neq i, \mathbf{b} \mathbf{v}_i > \beta.
 \end{aligned}$$

Thus $\mathbf{v}_i$ is a vertex, defined by the valid inequality $\mathbf{b} \mathbf{x} \leq \mathbf{b} \mathbf{v}_i$.

Finally we observe that a vertex $\mathbf{v}_i$ of P can never be written as a convex combination of points in $P \setminus \mathbf{v}_i$, which finishes the proof of both (i) and (ii). $\square$

Proposition 2.3. *Let $P \subseteq \mathbb{R}^d$ be a polytope, and $V := \text{vert}(P)$. Let F be a face of P .*

- (i) *The face F is a polytope, with $\text{vert}(F) = F \cap V$.*
- (ii) *Every intersection of faces of P is a face of P .*
- (iii) *The faces of F are exactly the faces of P that are contained in F .*
- (iv) *$F = P \cap \text{aff}(F)$.*

Proof. Let F be defined by the valid inequality $\mathbf{c}\mathbf{x} \leq c_0$.

For the first assertion of part (i), we see that F is a polytope from the characterization of polytopes as bounded intersections of halfspaces: F is the intersection of a polytope P with a polyhedron (hyperplane)

$$H := \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}.$$

Furthermore, we find that $F \subseteq \text{aff}(F) \subseteq H$, which proves (iv).

For the second assertion of (i), note that $\text{vert}(F) \supseteq F \cap V =: V_0$. For the converse inclusion, let $\mathbf{x} \in F$, so that $\mathbf{x}$ can be represented as $\mathbf{x} = V\mathbf{t}$, with $\mathbf{t} \geq 0$, $\mathbf{1}\mathbf{t} = 1$. We compute

$$c_0 = \mathbf{c}\mathbf{x} = \mathbf{c}(V\mathbf{t}) = (\mathbf{c}V)\mathbf{t} \leq c_0 \mathbf{1}\mathbf{t} = c_0,$$

thus $(\mathbf{c}\mathbf{v}_i - c_0)t_i = 0$ for all i . This implies that $t_i = 0$ for all i with $\mathbf{v}_i \notin V_0$, and thus $\mathbf{x} \in \text{conv}(V_0)$. From this we see $F = \text{conv}(V_0)$, and thus $\text{vert}(F) \subseteq V_0$ by Proposition 2.2(ii). This completes the proof of (i).

For (ii), let

$$F = P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}$$

and

$$G = P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{b}\mathbf{x} = b_0\}$$

for inequalities $\mathbf{c}\mathbf{x} \leq c_0$ and $\mathbf{b}\mathbf{x} \leq b_0$ that are valid for P . Then the inequality $(\mathbf{c} + \mathbf{b})\mathbf{x} \leq c_0 + b_0$ is valid for P , and

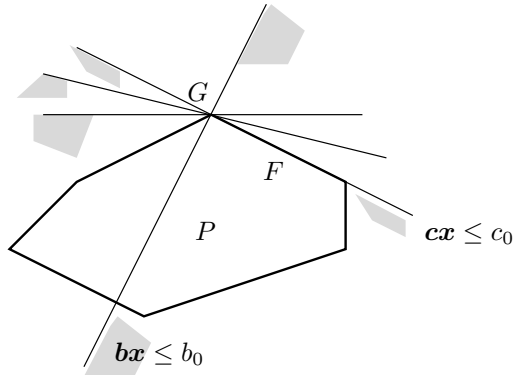
$$P \cap \{\mathbf{x} \in \mathbb{R}^d : (\mathbf{c} + \mathbf{b})\mathbf{x} = c_0 + b_0\} = F \cap G.$$

For (iii), if $G \subseteq F$ is a face of P , then it is a face of F as well. For the converse let $F = P \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{c}\mathbf{x} = c_0\}$ and $G = F \cap \{\mathbf{x} \in \mathbb{R}^d : \mathbf{b}\mathbf{x} = b_0\} \subseteq F$, where $\mathbf{c}\mathbf{x} \leq c_0$ is valid for P , and $\mathbf{b}\mathbf{x} \leq b_0$ is valid for F , but not necessarily for P .

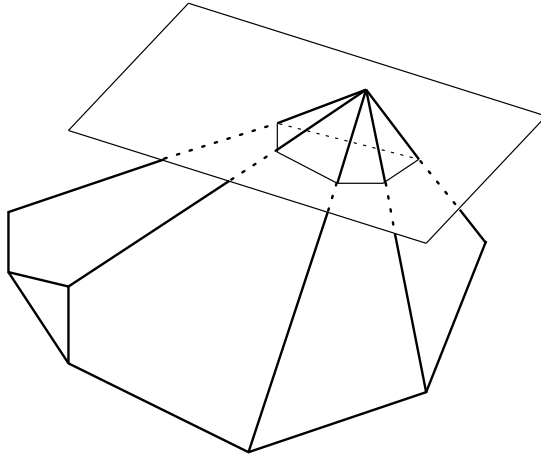
Let $V_0 := \text{vert}(F)$ as before, and $V_1 := V \setminus V_0$. We can assume that $F \neq P$, and thus $V_1 \neq \emptyset$. Now $(\mathbf{b} + \lambda\mathbf{c})\mathbf{x} \leq b_0 + \lambda c_0$ is valid for F , for every $\lambda \in \mathbb{R}$, and defines G as a face of F .

Now choose λ large enough to satisfy $\lambda > -\frac{b_0 - \mathbf{b}\mathbf{v}}{c_0 - \mathbf{c}\mathbf{v}}$ for all $\mathbf{v} \in V_1$. Then we get that $(\mathbf{b} + \lambda\mathbf{c})\mathbf{x} \leq b_0 + \lambda c_0$ is valid with strict inequality for all $\mathbf{v} \in V_1$.

We conclude that G is a face of P . (The sketch below might tell you what was “really going on” during this algebraic manipulation.) $\square$



We need another construction: the “vertex figure” obtained by cutting a polytope by a hyperplane that cuts off a single vertex.



For this, we consider a polytope P with $V = \text{vert}(P)$, and a vertex $\mathbf{v} \in V$. Let $\mathbf{c}\mathbf{x} \leq c_0$ be a valid inequality with

$$\{\mathbf{v}\} = P \cap \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_0\}.$$

Furthermore, we choose some $c_1 < c_0$ with $\mathbf{c}\mathbf{v}' < c_1$ for all $\mathbf{v}' \in \text{vert}(P) \setminus \mathbf{v}$. Then we define a *vertex figure* of P at $\mathbf{v}$ as the polytope

$$P/\mathbf{v} := P \cap \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_1\}.$$

Note that the construction of $P/\mathbf{v}$ depends on the choice of c_1 and of the inequality $\mathbf{c}\mathbf{x} \leq c_0$; however, the following result shows that the combinatorial type of $P/\mathbf{v}$ is independent of this.

Proposition 2.4. *There is a bijection between the k -dimensional faces of P that contain $\mathbf{v}$, and the $(k-1)$ -dimensional faces of $P/\mathbf{v}$, given by*

$$\begin{aligned} \pi : F &\longmapsto F \cap \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_1\}, \\ \sigma : P \cap \text{aff}(\{\mathbf{v}\} \cup F') &\longleftarrow F'. \end{aligned}$$

Proof. Denote the “cutting hyperplane” by $H := \{\mathbf{x} : \mathbf{c}\mathbf{x} = c_1\}$.

The map π is well defined: let $F = P \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$; then we get $F \cap H = (P \cap H) \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$, where $\mathbf{b}\mathbf{x} \leq b_0$ is valid for P and thus also for $P/\mathbf{v}$.

To see that σ is well defined, let $F' = (P \cap H) \cap \{\mathbf{x} : \mathbf{b}\mathbf{x} = b_0\}$, where $\mathbf{b}\mathbf{x} \leq b_0$ is valid for $P/\mathbf{v}$. Then $(\mathbf{b} + \lambda\mathbf{c})\mathbf{x} \leq b_0 + \lambda c_1$ is valid for $P/\mathbf{v}$, for all $\lambda \in \mathbb{R}$. Now a simple computation shows that for

$$\lambda_0 := \frac{b_0 - \mathbf{b}\mathbf{v}}{c_0 - c_1},$$

the inequality is valid on P , with equality for $\mathbf{v}$. In fact, if we consider $\mathbf{v}' \in V \setminus \mathbf{v}$, then we know $\mathbf{c}\mathbf{v}' < c_1$ and $\mathbf{c}\mathbf{v} = c_0 > c_1$, so

$$\mathbf{v}'' := \frac{(\mathbf{c}\mathbf{v} - c_1)\mathbf{v}' + (c_1 - \mathbf{c}\mathbf{v}')\mathbf{v}}{\mathbf{c}\mathbf{v} - \mathbf{c}\mathbf{v}'} \in P \cap H = P/\mathbf{v}.$$

Now $\mathbf{v}''$ is a convex combination of $\mathbf{v}$ and $\mathbf{v}'$, and from $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v}'' \leq b_0 + \lambda_0 c_1$ and $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v} = b_0 + \lambda_0 c_1$ we get $(\mathbf{b} + \lambda_0\mathbf{c})\mathbf{v}' \leq b_0 + \lambda_0 c_1$.

Now we check that the maps σ and π are inverses of each other: we compute

$$\pi \circ \sigma(F') = H \cap P \cap \text{aff}(\{\mathbf{v}\} \cup F') = P \cap H \cap \text{aff}(F') = P/\mathbf{v} \cap \text{aff}(F') = F',$$

where the last equality is from Proposition 2.3(iv). Similarly,

$$\sigma \circ \pi(F) = P \cap \text{aff}(\{\mathbf{v}\} \cup (F \cap H)) \stackrel{*}{=} P \cap \text{aff}(F) = F,$$

where for the equality $*$ we use that every vertex $\mathbf{v}'$ of F can be obtained as an affine combination of $\mathbf{v}$ and a point $\mathbf{v}'' \in F \cap H$.

Finally we observe that if F' is a face of $P/\mathbf{v}$ of dimension $k-1$, then the associated face F of P has dimension k , since its affine hull is $\text{aff}(F) = \text{aff}(F' \cup \{\mathbf{v}\})$, where $\mathbf{v} \notin \text{aff}(F')$ by construction. $\square$

2.2 The Face Lattice

In this section we translate some of our results on polytopes into purely combinatorial statements. For this we need some terminology about finite *partially ordered sets* (“posets,” for short). We refer to Stanley’s book [517, Sects. 3.1–3.3] for more information on that subject and its ramifications. For simplicity, and to unify terminology, we define the key concepts here.